

Phenotypic synergy testing of ceftazidime–avibactam with aztreonam in a university hospital having high number of metalloβ-lactamase producing bacteria

Chinmoy Sahu, Sourav Pal, Sangram Singh Patel, Sanjay Singh, Mohan Gurjar & Ujjala Ghoshal

To cite this article: Chinmoy Sahu, Sourav Pal, Sangram Singh Patel, Sanjay Singh, Mohan Gurjar & Ujjala Ghoshal (2020): Phenotypic synergy testing of ceftazidime–avibactam with aztreonam in a university hospital having high number of metalloβ-lactamase producing bacteria, Infectious Diseases, DOI: [10.1080/23744235.2020.1788224](https://doi.org/10.1080/23744235.2020.1788224)

To link to this article: <https://doi.org/10.1080/23744235.2020.1788224>



Published online: 06 Jul 2020.



Submit your article to this journal [↗](#)



Article views: 23



View related articles [↗](#)



View Crossmark data [↗](#)

ORIGINAL ARTICLE



Phenotypic synergy testing of ceftazidime–avibactam with aztreonam in a university hospital having high number of metalloβ-lactamase producing bacteria

Chinmoy Sahu^a, Sourav Pal^a, Sangram Singh Patel^a, Sanjay Singh^a, Mohan Gurjar^b  and Ujjala Ghoshal^a

^aDepartment of Microbiology, Sanjay Gandhi Post Graduate Institute of Medical Sciences, Lucknow, India; ^bDepartment of Critical Care Medicine, Sanjay Gandhi Post Graduate Institute of Medical Sciences, Lucknow, India

ABSTRACT

Background: Ceftazidime–avibactam combination with aztreonam and role of rapid synergy reporting has not been widely evaluated. Also the synergy correlation with various β-lactamases has not been widely studied.

Methods: We studied phenotypic synergy testings and molecular detection of β-lactamases in our university hospital where we have large number of metalloβ-lactamase producing bacteria. We tested two phenotypic synergy methods for ceftazidime–avibactam with aztreonam (Disc-E strip method, E strip-Agar method) for rapid reporting to clinicians (153 isolates). The treatment (colistin, ceftazidime–avibactam, ceftazidime–avibactam with aztreonam) was guided as indicated in the synergy testings. The resistance genes in bacteria were identified by polymerase chain reaction (PCR) and correlated with synergy results.

Results: The highest synergy was seen in *Klebsiella pneumoniae* by Disc-E strip and E strip-Agar method (86% and 84% respectively). About 70% of *Pseudomonas aeruginosa* and 29% of *Escherichia coli* showed synergy. Molecular methods revealed multiple resistance gene combinations and *bla*_{NDM} (96%) was predominant gene in isolates showing synergy. Among isolates that were sensitive to ceftazidime–avibactam, the predominant genes were *bla*_{OXA-48} and *bla*_{IMP}. Rapid laboratory reporting led to proper utilization of antibiotic combinations.

Conclusions: Ceftazidime–avibactam and aztreonam rapid synergy testing will be highly beneficial in treatment of infections by metalloβ-lactamase producing resistant bacteria, especially *K. pneumoniae* and *P. aeruginosa*.

KEYWORDS

Synergy
phenotypic
ceftazidime–avibactam
aztreonam
clinical

ARTICLE HISTORY

Received 20 April 2020
Revised 10 June 2020
Accepted 22 June 2020

CONTACT

Chinmoy Sahu
 sahu.chinmoy@gmail.com
 Department of Microbiology, Sanjay Gandhi
Post Graduate Institute of Medical Sciences,
Lucknow 226014, India

Introduction

In tertiary health care centres, Multi Drug Resistance (MDR) is common in ESKAPE pathogens [1]. Although we have some antibiotic options against Gram-positive pathogens, there are a few antibiotic options left for Gram-negative pathogens. Only colistin, tigecycline and fosfomycin are some of the options but toxicity of these agents (especially colistin) sometimes preclude their use in critically ill patients.

Beta lactam (especially cephalosporins like ceftazidime and ceftriaxone) agents are excellent against Gram negative pathogens because of their broad-spectrum activity, safety and well tolerability. However, their role in MDR pathogens are diminishing due to very high prevalence of enzymes like extended spectrum beta lactamases (ESBLs), Metallo betalactamses (MBLS), AmpC beta lactamses and OXA-48 producing pathogens. In many centres, MDR pathogens have the combinations of these enzymes, so that therapeutic options are very limited [2].

Ceftazidime–avibactam is combination of cephalosporin (ceftazidime) and betalactamase inhibitor (avibactam). Avibactam inhibits ESBLs, AmpC betalactamses, OXA-48 and serine carbapenemases like *Klebsiella pneumoniae* carbapenemase (KPC). However, it can't inhibit metallobetalactamses like New Delhi Metallo Betalactamses (NDM) [3]. Aztreonam is a monobactam antibiotic. The metallobetalactamses like NDM can't destroy aztreonam. But ESBLs and other enzymes inactivate the antibiotic [4].

So, the combination of Ceftazidime-avibactam and aztreonam can act against all betalactamase producing Gram-negative pathogens [5]. If effective, it can be an excellent option in high ESBL, MBL prevalent health centres. Combination of ceftazidime–avibactam with colistin can be a choice in Pan Drug Resistant (PDR) patients and also it will help in reducing the dose of colistin.

MDR Gram-negative bacteria isolates typically contain multiple resistance mechanisms and genes [6]. So, it becomes imperative to use multiple antibiotics in combination. It is prudent to know the synergistic, additive or antagonistic activity of these antibiotics. It also helps the clinicians in choosing the right combination of antibiotics. Recommended methods of antibiotic synergy testing like checkerboard broth microdilution method is time consuming, exhaustive, costly and impractical in day to day reporting to clinicians. So, rapid phenotypic synergy testings are required to recommend synergy of

ceftazidime-avibactam and aztreonam. Also, the clinical response to treatment by the antibiotic combination is important to study.

So, we decided to study the efficacy of ceftazidime–avibactam; ceftazidime–avibactam with aztreonam by two laboratory phenotypic methods in clinical bacterial isolates. We studied resistance genes in these isolates and also noted the clinical response in some patients.

Materials and methods

The observational experimental study was conducted in Department of Microbiology and Department of Critical Care Medicine of our tertiary health care university hospital. Bacterial isolates cultured from clinical samples (blood, urine, sputum, pus) from critical care patients were included in the study. All the patients were of various hospital acquired infections like catheter related blood stream infections (CRBSI), catheter-associated urinary tract infection (CAUTI), and hospital-acquired pneumonia (HAP) and surgical site infections (SSI). Ethical clearance was taken from Institute Ethical Committee.

Bacterial identification and antibiotic sensitivity tests

All Gram-negative bacteria were tested for sensitivity to first line antibiotics. MDR bacteria (isolates resistant to three or more class of antibiotics) were detected by Kirby–Bauer disc diffusion method. Identification of the bacterial isolates was confirmed by Matrix Assisted Laser Desorption Ionization Mass Spectrometry (MALDI-TOF MS) method (VITEK MS, Biomerieux, France). Minimum inhibitory concentration (MIC) of various antibiotics was confirmed by automated bacterial identification and sensitivity methods (Phoenix, BD, Tempe, AZ). For colistin, the MIC was confirmed by broth microdilution method.

Testing of ceftazidime–avibactam; ceftazidime–avibactam and aztreonam synergy was done by different phenotypic methods.

- MIC of ceftazidime–avibactam was detected by E-test strip (Pfizer, New York, NY) method containing ceftazidime (0.01–256 µg/ml)–avibactam (4 µg/ml) on Müller Hinton Agar for all MDR bacterial isolates. Cut-off sensitivity was taken as per CLSI 2019 guidelines (8 µg/mL) [7].
- Disc-E strip synergy testing. Mueller–Hinton agar plates (150 mm diameter) were inoculated with suspensions of study isolates grown to an optic density of 0.5 McFarland units. E test strip of

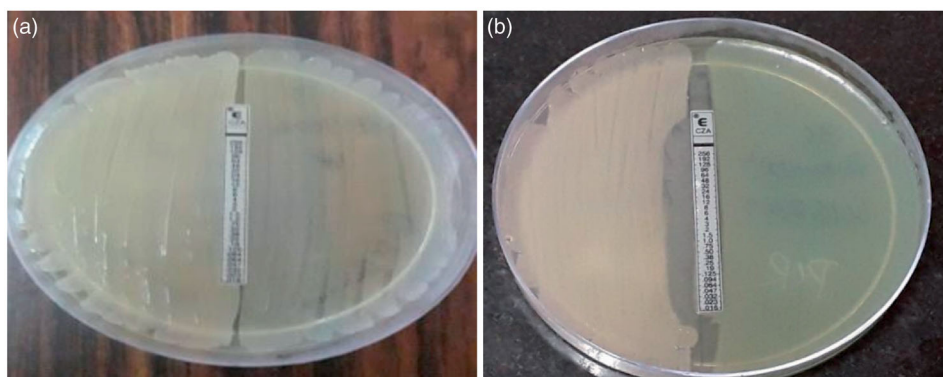


Figure 1. (a) Ceftazidime-avibactam E strip MIC testing. (b) E strip-Agar synergy testing. The same two bacterial isolates on each plate.

ceftazidime-avibactam was put on the plate. Then, aztreonam disc (30 µg) (Thermo Fisher Scientific, Mumbai, India) was put near the E test strip such that centre of disc remained parallel to the cut-off sensitivity mark of ceftazidime-avibactam (8 µg/mL). The distance between the strip and disc was 30 mm (centre to centre). The plate was incubated for 18 h. The reduction of MIC by ≥ 3 -fold was reported as synergy.

- E strip-Agar synergy testing: Müller Hinton Agar containing aztreonam antibiotic powder (4 µg/ml) was prepared (Merck, Mumbai, India). On this, E test strip of ceftazidime-avibactam was put. The plate was incubated for 18 h. Cut-off sensitivity was taken as per CLSI 2019 guidelines (8 µg/ml) [7]. The reduction of MIC by ≥ 3 -fold was reported as synergy.
- Study of molecular resistance mechanisms: Genotypic detection of betalactamases (ESBL, AmpC) [8], carbapenemase (OXA, KPC) [9], metallo-carbapenemase (IMP, VIM) [10] and NDM [9] by Polymerase Chain Reaction (PCR) was done. Colistin resistance (mcr-1) [11] was detected by PCR.
- Patient follow up and clinical scenario: From the start, the clinical details and antibiotic history were noted. The patients were followed up for antibiotic response, escalation or de-escalation. Ceftazidime-avibactam (2.5 gm 8 hourly for minimum 5 days) with aztreonam (2 gm 8 hourly for minimum 5 days) were given to some patients where the isolated bacteria were sensitive to the antibiotic synergy *in vitro*.

Results

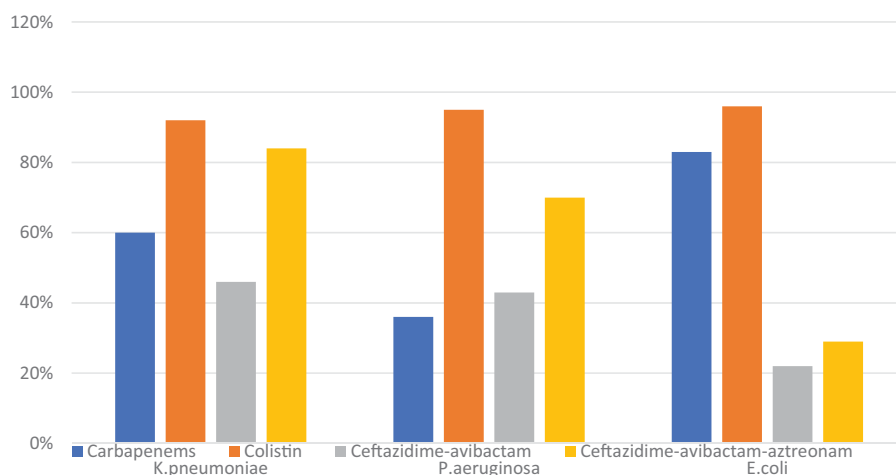
From the critical care centre, 153 MDR Gram-negative bacterial isolates were obtained. Among them 24%, 38%, 22% isolates were from blood, respiratory samples



Figure 2. Disc-E test synergy testing of ceftazidime-avibactam and aztreonam.

and urinary samples respectively. The most common bacterial species isolated was *K. pneumoniae* (48) followed by *Pseudomonas aeruginosa* (36), *Escherichia coli* (29) and *Acinetobacter spp* (15). *Klebsiella pneumoniae* was most common from CRBSI and HAP cases. *Escherichia coli* was most common from CAUTI and SSI cases. The mean age among CRBSI was less (47.1 years) than CAUTI (52.3 years) and HAP (57.7 years) cases.

Synergy testing was done for three most common bacterial isolates (*K. pneumoniae*, *P. aeruginosa* and *E. coli*) (Figures 1 and 2). Among 48 *K. pneumoniae* isolates, 29 (60%) were sensitive to carbapenems, 44 (92%) were sensitive to colistin. 23 (46%) were sensitive to ceftazidime-avibactam. Forty-two (84%) of the isolates showed synergy of ceftazidime-avibactam and aztreonam by E strip-Agar method and 41 (82%) of the isolates showed synergy by Disc-E strip method (Graph 1 and Table 1).



Graph 1. Comparison of sensitivity to various antibiotics including ceftazidime–avibactam–aztreonam.

Table 1. Comparison of the phenotypic synergy testing methods.

Bacteria	E strip-plate method	E strip-disc method
<i>K. pneumoniae</i> (n = 48)	42 (84%)	41 (82%)
<i>P. aeruginosa</i> (n = 36)	24 (70%)	24 (70%)
<i>E. coli</i> (n = 29)	9 (29%)	9 (29%)

Thirty-six MDR *P. aeruginosa* were isolated. Among them 14 (36%) were sensitive to carbapenems, and 34 (95%) were sensitive to colistin. Fifteen (43%) were sensitive to ceftazidime–avibactam and 24 (70%) showed synergy by both E strip-Agar method and Disc-E strip method (Graph 1 and Table 1).

Among the 29 *E. coli* isolates, 24 (83%) were sensitive to carbapenems and 28 (97%) were sensitive to colistin. Seven (22%) were sensitive to ceftazidime–avibactam and only 9 (29%) showed synergy by both E strip-Agar method and Disc-E strip method (Graph 1 and Table 1).

We tested for different molecular mechanisms in all the bacterial isolates. PCR and sequencing were done to detect the resistant genes. Among isolates that were sensitive to ceftazidime–avibactam, the predominant genes were *bla*_{OXA-48} and *bla*_{IMP} (Table 2). Among the isolates that were resistant to ceftazidime–avibactam, but showed synergy to ceftazidime–avibactam and aztreonam, the predominant genes were *bla*_{NDM} and its variants. In isolates that were resistant to the combination, *bla*_{NDM} and its variants were also the most common genes.

Antibiotic therapies were altered in patients after sensitivity reports. Ceftazidime–avibactam with aztreonam was given in five patients out of which four patients responded to treatment with clinical improvement (Table 3). MDR bacteria (three *K. pneumoniae* and one *P.*

Table 2. Resistance genes in various groups of sensitivity.

Isolates	Resistance genes
Sensitive to ceftazidime–avibactam (n = 45)	<i>bla</i> _{TEM} (86%) <i>bla</i> _{SHV} (74%) <i>bla</i> _{CTX-M} (57%) <i>bla</i> _{AmpC} (17%) <i>bla</i> _{OXA-48} (33%) <i>bla</i> _{KPC} (12%) <i>bla</i> _{NDM} (8%) <i>bla</i> _{NDM} (6%) <i>bla</i> _{VIM} (2%)
Sensitive to ceftazidime–avibactam and aztreonam combination only – showing synergy (n = 30)	<i>bla</i> _{TEM} (73%) <i>bla</i> _{SHV} (65%) <i>bla</i> _{CTX-M} (45%) <i>bla</i> _{AmpC} (29%) <i>bla</i> _{OXA-48} (37%) <i>bla</i> _{KPC} (17%) <i>bla</i> _{NDM} (96%) <i>bla</i> _{IMP} (63%) <i>bla</i> _{VIM} (43%)
Resistant to ceftazidime–avibactam and aztreonam (n = 38)	<i>bla</i> _{TEM} (70%) <i>bla</i> _{SHV} (65%) <i>bla</i> _{CTX-M} (59%) <i>bla</i> _{AmpC} (19%) <i>bla</i> _{OXA-48} (30%) <i>bla</i> _{KPC} (7%) <i>bla</i> _{NDM} (85%) <i>bla</i> _{NDM} (15%) <i>bla</i> _{VIM} (19%)
Resistant to colistin (n = 7)	<i>mcr-1</i> (42%)

aeruginosa) were isolated in all these five patients and all were resistant to colistin. All the bacteria isolated from these patients were positive for synergy testing. One patient with CAUTI due to *E. coli* did not respond to treatment.

Discussion

MDR bacteria cause significant problem in choosing the right antibiotic. Multiple factors contribute to the problem. Majority of MDR, carbapenem-resistant bacteria are isolated from inpatients having hospital acquired

Table 3. Clinical response to ceftazidime–avibactam and ceftazidime–avibactam–aztreonam.

Serial number	Age	Sex	Clinical condition	Organism	Synergy testing	Outcome
1	55	F	HAP	<i>K. pneumoniae</i>	Positive	Responded
2	68	M	HAP	<i>K. pneumoniae</i>	Positive	Responded
3	70	M	CRBSI	<i>P. aeruginosa</i>	Positive	Responded
4	56	M	HAP	<i>K. pneumoniae</i>	Positive	Responded
5	71	M	CAUTI	<i>E. coli</i>	Positive	Not responded

infections. They usually have multiple co-morbidities, immunosuppression, prior antibiotic use history etc. Resistance to safer drugs like carbapenems often pose challenge in antibiotic therapy especially in hospitals having high load of mellaobetalactamse-producing bacteria. Colistin, the last line of defence often can't be used in these patients due to toxicity. Some bacteria like *Stenotrophomonas maltophilia*, *Chryseobacterium indologenes* and *Proteus* spp. are intrinsically resistant to colistin. So, in many cases, the clinicians look for other alternative antibiotics.

Combined antibiotic strategy helps in overcoming antibiotic resistance and improving efficacy. Although the rationale of combining ceftazidime–avibactam with aztreonam is theoretically conclusive, the practical solution should be tested in individual situation. Microbroth dilution chequerboard assay and time-kill assays are gold standards of antibiotic synergy testings [12]. However these methods are time consuming, labour intensive and less suitable for day to day reporting. Various studies have reported use of disc overlapping method, E-strip synergy and agar dilution methods to report synergy. A study compared E-test synergy, checker board method and time-kill assay on 32 *Acinetobacter baumannii* isolates for various antibiotics like imipenem, tigecycline, colistin, amikacin, etc. [13]. It found E-test agar method more comparable to checkerboard and time-kill assays in detection of synergy. Two phenotypic synergy tests (E strip–E strip and E strip–agar) were evaluated by Wenzler et al. in seven MDR bacterial isolates for ceftazidime, ceftazidime–avibactam and aztreonam [14]. All isolates were resistant to ceftazidime and almost all were resistant to aztreonam. E strip–E strip MIC method showed synergy in 71% isolates and E strip–agar method to 86% isolates. A study from India reported the MICs of ceftazidime–avibactam reduced > 8-fold in 12 isolates of carbapenem resistant *Klebsiella pneumoniae* [15]. Only 2 (18%) out of 12 isolates were susceptible to ceftazidime–avibactam alone. However, the authors used checkerboard assay to determine the synergy without clinical correlation. In our study, we found similar synergy by both our phenotypic methods (Disc-E strip method and E strip–Agar method).

Ceftazidime–avibactam was sensitive in approximately 45–50% of isolates and even less so in *E. coli* isolates. However, ceftazidime–avibactam and aztreonam synergy was seen almost 1.5 times to that (70–84% of the isolates) especially in *K. pneumoniae* and *P. aeruginosa* isolates. Similar phenotypic synergy testing (disc overlapping and agar dilution method) and correlation with neutropenic mouse model has been reported [5].

Sensitivity to ceftazidime–avibactam as well as synergy with aztreonam was found predominantly in *K. pneumoniae* followed by *P. aeruginosa* in our study. Both sensitivity and synergy were least for *E. coli*. Overexpression of efflux pumps like MexAB–OprM in *P. aeruginosa* can lead to extrusion of ceftazidime from cytoplasm and failure of synergy [16]. Similarly altered penicillin binding protein (PBP) can lead to reduced efficacy of ceftazidime and lack of synergy. A study was done in India on *E. coli* isolates showing high MICs (2–32 µg/mL) to aztreonam/avibactam [17]. Majority of the isolates showed insertions of four amino acids (YRIN and YRIK) in PBP of both NDM and OXA48 producing isolates.

Molecular resistance mechanisms and epidemiology play important role in deciding antibiotic therapy and antibiotic policy. High prevalence of ESBL (CTX-M, SHV); MBL (IMP, NDM); serine carbapenemase (KPC) and mcr-1 in bacterial isolates can lead to multiple drug resistance and treatment limitation. Ceftazidime–avibactam can work on Ambler class A, C and D betalactamases including KPC and not on class B metallobetalactamses [3]. So, molecular epidemiology of a particular health set up is important for proper antibiotic choice (ceftazidime–avibactam alone or with aztreonam). We found high prevalence of TEM, SHV, OXA-48 in isolates showing sensitivity to ceftazidime–avibactam. High prevalence of NDM and IMP were seen in isolates showing resistance to ceftazidime–avibactam but showing synergy with aztreonam. Similar synergy and molecular mechanism correlation were also found by many authors [5,15,18].

Ceftazidime–avibactam has been studied in many randomized controlled trials for treatment of complicated UTI, HAP and intra-abdominal infections (IAI) [19–21]. All the studies found non-inferiority of

ceftazidime-avibactam over comparators. However, clinical studies of combination of ceftazidime-avibactam with aztreonam are less. In one case report, patient of CRBSI due to NDM and OXA-48 producing *K. pneumoniae* did not respond to colistin and amikacin [22]. However E-strip-E-strip testing indicated synergy between ceftazidime-avibactam and aztreonam. The patient was given combination therapy of both the antibiotics and responded. The same report also mentioned another patient of pneumonia due to NDM and AmpC producing *P. aeruginosa*. He recovered quickly after combination therapy as indicated by synergy testing. Another case report mentioned successful treatment of hip arthroplasty infection by *Enterobacter cloacae* by the antibiotic combination [5]. This synergy was also reported by disc:disc overlapping method. In our cases, four patients responded to the combination treatment. All four bacteria causing the infections were positive for synergy testing and all were resistant to colistin. The rapid reporting of synergy helped the clinicians to take decisions for antibiotic selection.

So, this study was the first to observe large number clinical bacterial isolates from hospital acquired infections. It indicated utility of rapid phenotypic synergy testing of antibiotic combination to guide the clinicians. This study emphasizes the role of optimum utilization of phenotypic testing and molecular epidemiology in treatment of MDR bacteria especially *K. pneumoniae* and *P. aeruginosa*. In this context, randomized controlled trials are required to evaluate this antibiotic combination of ceftazidime-avibactam and aztreonam.

Acknowledgement

The authors thank Pfizer, New York, NY, for supply of E strips.

Disclosure statement

The authors declare no conflicts of interest.

ORCID

Mohan Gurjar  <http://orcid.org/0000-0002-8489-0324>

References

- [1] Santajit S, Indrawattana N. Mechanisms of antimicrobial resistance in ESKAPE pathogens. *Biomed Res Int*. 2016; 2016:2475067.
- [2] Mohanty S, Gajanand M, Gaind R. Identification of carbapenemase-mediated resistance among *Enterobacteriaceae* bloodstream isolates: a molecular study from India. *Indian J Med Microbiol*. 2017;35(3):421–425.
- [3] Shirley M. Ceftazidime-avibactam: a review in the treatment of serious Gram-negative bacterial infections. *Drugs*. 2018; 78(6):675–692.
- [4] Rawat D, Nair D. Extended-spectrum β -lactamases in Gram negative bacteria. *J Glob Infect Dis*. 2010;2(3):263–274.
- [5] Marshall S, Hujer AM, Rojas LJ, et al. Can Ceftazidime-Avibactam and Aztreonam overcome β -lactam resistance conferred by metallo- β -lactamases in Enterobacteriaceae? *Antimicrob Agents Chemother*. 2017; 61(4):e02243.
- [6] Exner M, Bhattacharya S, Christiansen B, et al. Antibiotic resistance: what is so special about multidrug-resistant Gram-negative bacteria? *GMS Hyg Infect Control*. 2017;12:Doc05.
- [7] Clinical and Laboratory Standards Institute. M100 Performance standards for antimicrobial susceptibility testing. 29th ed. Wayne (PA): Clinical and Laboratory Standards Institute; 2010.
- [8] van den Bijllaardt W, Janssens MM, Buiting AG, et al. Extended-spectrum β -lactamase (ESBL) polymerase chain reaction assay on rectal swabs and enrichment broth for detection of ESBL carriage. *J Hosp Infect*. 2018;98(3):264–269.
- [9] Lee TD, Adie K, McNabb A, et al. Rapid detection of KPC, NDM, and OXA-48-like carbapenemases by real-time PCR from rectal Swab surveillance samples. *J Clin Microbiol*. 2015;53(8):2731–2733.
- [10] Manoharan A, Chatterjee S, Mathai D, SARI Study Group. Detection and characterization of metallo beta lactamases producing *Pseudomonas aeruginosa*. *Indian J Med Microbiol*. 2010;28(3):241–244.
- [11] Rolain JM, Kempf M, Leangapichart T, et al. Plasmid-mediated *mcr-1* gene in colistin-resistant clinical isolates of *Klebsiella pneumoniae* in France and Laos. *Antimicrob Agents Chemother*. 2016; 60(11):6994–6995.
- [12] Laishram S, Pragasam A, Bakthavatchalam YD, et al. An update on technical, interpretative and clinical relevance of antimicrobial synergy testing methodologies. *Indian J Med Microbiol*. 2017;35(4):445–468.
- [13] Sopirala MM, Mangino JE, Gebreyes WA, et al. Synergy testing by Etest, microdilution checkerboard, and time-kill methods for pan-drug-resistant *Acinetobacter baumannii*. *Antimicrob Agents Chemother*. 2010;54(11):4678–4683.
- [14] Wenzler E, Deraedt MF, Harrington AT, et al. Synergistic activity of ceftazidime-avibactam and aztreonam against serine and metallo- β -lactamase-producing gram-negative pathogens. *Diagn Microbiol Infect Dis*. 2017;88(4):352–354.
- [15] Pragasam AK, Veeraraghavan B, Shankar BA, et al. Will ceftazidime/avibactam plus aztreonam be effective for NDM and OXA-48-like producing organisms: lessons learnt from *in vitro* study. *Indian J Med Microbiol*. 2019;37(1):34–41.
- [16] Masuda N, Sakagawa E, Ohya S, et al. Substrate specificities of MexAB-OprM, MexCD-OprJ, and MexXY-oprM efflux pumps in *Pseudomonas aeruginosa*. *Antimicrob Agents Chemother*. 2000;44(12):3322–3327.
- [17] Periasamy H, Joshi P, Palwe S, et al. High prevalence of *Escherichia coli* clinical isolates in India harbouring four amino acid inserts in PBP3 adversely impacting activity of

- aztreonam/avibactam. *J Antimicrob Chemother.* 2020;75(6):1650–1651.
- [18] Sader HS, Mendes RE, Pfaller MA, et al. Antimicrobial activities of Aztreonam–Avibactam and comparator agents against contemporary (2016) clinical *Enterobacteriaceae* isolates. *Antimicrob Agents Chemother.* 2017; 62(1):e01856.
- [19] Carmeli Y, Armstrong J, Laud PJ, et al. Ceftazidime-avibactam or best available therapy in patients with ceftazidime-resistant *Enterobacteriaceae* and *Pseudomonas aeruginosa* complicated urinary tract infections or complicated intra-abdominal infections (REPRISE): a randomised, pathogen-directed, phase 3 study. *Lancet Infect Dis.* 2016;16(6):661–673.
- [20] Stone GG, Newell P, Gasink LB, et al. Clinical activity of ceftazidime/avibactam against MDR *Enterobacteriaceae* and *Pseudomonas aeruginosa*: pooled data from the ceftazidime/avibactam Phase III clinical trial programme. *J Antimicrob Chemother.* 2018;73(9):2519–2523.
- [21] Torres A, Zhong N, Pacht J, et al. Ceftazidime-avibactam versus meropenem in nosocomial pneumonia, including ventilator-associated pneumonia (REPROVE); a randomized, double-blind, phase 3 non-inferiority trial. *Lancet Infect Dis.* 2018;18(3):285–295.
- [22] Davido B, Fellous L, Lawrence C, et al. Ceftazidime–Avibactam and Aztreonam, an interesting strategy to overcome β -lactam resistance conferred by metallo- β -lactamases in *Enterobacteriaceae* and *Pseudomonas aeruginosa*. *Antimicrob Agents Chemother.* 2017;61(9):e01008.